

WILLIAM R. JOHNSON

SOFTWARE ENGINEER

wrjstuff@gmail.com
Atlanta, GA
CooCooCaCha

EXPERIENCE

CancerIQ
2022 - present

Principal Software Engineer
Promoted from Senior Software Engineer in 2026.

Built apps for cancer prevention and early detection using Rails, React, Rust, Elixir, Node.js, and Google Cloud.

- Led the genogram team in rewriting our visual family tree builder in React and refactoring the Rails backend to support new features.
- Researched and developed a deterministic, recursive family tree layout algorithm that preserves paternal/maternal orientation and keeps twins adjacent, replacing our previous Sugiyama-based layout.
- Built a custom query language using parser combinators, enabling product managers to target content based on patient data.
- Worked with a co-founder to standardize an Excel format for patient data and built a Ruby import pipeline using pre-generated UUIDs and transactions to prevent partial patient imports and resume safely after failures. Tested with 1M+ records and 100+ columns.
- Led improvements to our AI-assisted development workflow by grouping apps and services in one folder and writing an AGENTS.md to give AI tools codebase context and development guidelines.

World 50
2018 - 2022

Principal Software Engineer
Promoted from Senior Software Engineer in 2021.

Manager of 5 engineers, and full-stack software engineer dedicated to rebuilding our company's software platform on react-native and nodejs.

- Mentored engineers on my team, performed 1-on-1's, and wrote yearly performance reviews.
- Responsible for architecting, coding, and maintaining core systems used by my team.
- Wrote engineering documentation and introduced architecture decision records (ADRs) to our documentation.
- Advocated for restructing our teams to a product teams-like structure. This was eventually implemented.
- Rebuilt our legacy application on a modern tech stack using React-Native. The application runs on web, iOS, and Android with the majority of the code shared between all three platforms.
- Rebuilt our API using GraphQL. Additionally, we avoid N+1 query problems by using Facebook's Dataloader library.
- Setup infrastructure on AWS, Google Cloud and CircleCI including databases, servers, and build pipelines.
- Architected prototypes for a new navigation system and a better architecture for web. Both were accepted and implemented.

TSYS
2017 - 2018

Senior Software Engineer
ReactJS and front-end specialist.

- Gave presentations and introduced new React techniques such as Fragments and Render-Props (hooks were released after I left, but we would have used those instead).
- Completely rebuilt the UI for my product.
- Built a dynamically generated page flow system. Multi-page flows were generated from data returned from our REST API.
- Regularly contributed to our core React frameworks.
- Mentored and trained junior developers on React.

Koble
2016 - 2017

Co-Founder / Lead Engineer
Made all of the software architecture decisions. Wrote all of the code. Responded to outages and server issues.

- Developed a homepage using React and a prototype app using React-Native.
- Setup and maintained infrastructure using Google Cloud, including a Kubernetes cluster.
- Built a service to automatically retrieve and update LetsEncrypt certificates for TLS/SSL encryption.
- Wrote an audit-log service that listened for changes in Firebase and inserted a change summary into Google BigQuery.

Salesloft
2015 - 2016

Full-Stack Software Engineer
Designed, developed, and deployed features for the Salesloft SAAS application. This touched upon everything from database design and migrations to front-end UX design and development.

- Developed a new QA system using Docker. Different versions of our application could be tested simultaneously on the same server cluster.
- Learned Ruby, Ruby on Rails, and Angular on the job. Started contributing quickly.

EDUCATION

Georgia Institute of Technology
Bachelor's Degree - Computer Science
2007 - 2012

SKILLS

- Javascript
- Typecript
- React
- React-Native
- GraphQL
- Rust
- NodeJS
- SQL
- Git
- AWS and Google Cloud
- HTML
- CSS
- Rails
- Python
- Ruby